Blaque Baux — Financial Methods

A risk-premium harvesting platform: the mathematics, honestly.

Version 1.0 · 2026-07-30 · reference implementation in Julia (execution + strategy) with an archived Python research track.

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0. Abstract

Blaque Baux is a systematic, governed trading platform. Its live strategy — the **spine** — does **not** attempt to predict returns. It harvests **risk premia** (diversification, trend, term/asset premia) and manages risk with a stateful daily process, then routes every order through a governed execution layer that enforces hard invariants. This document specifies the financial mathematics as implemented, and states plainly what is validated, what is research, and what did **not** work.

The central empirical result that shaped the platform: on the instruments and horizons tested, **there was no exploitable predictive edge** (information coefficient ≈ 0; see §7). The edge that *does* exist and survives out-of-sample is in the **risk structure** — diversification and risk-timing — not in return forecasting. Everything below follows from that finding.

> **Not investment advice.** This is an engineering/quantitative reference. All performance figures are historical backtests or paper-trading results, net of modeled costs; they are not a promise of future results. Trading involves risk of loss.

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1. Notation and data

Let there be N assets and discrete time steps $t = 1, \dots, T$ (daily bars unless stated).

- **Prices** $P_{t,i}$ (split/dividend-adjusted, i.e. total-return closes).
- **Simple returns**

$$r_{t,i} = \frac{P_{t,i}}{P_{t-1,i}} - 1.$$

- $R \in \mathbb{R}^{T \times N}$ is the return matrix (rows = time, columns = assets). This $T \times N$ convention is used throughout the code.
- $w_t \in \mathbb{R}^N$ is the target weight vector at time t (fractions of capital; may be negative for shorts; $\sum_i |w_{t,i}|$ is **gross** exposure, $\sum_i w_{t,i}$ is **net**).
- Annualization uses **ppy** periods per year (252 for daily).

Reference universe (US ETFs, chosen for long history and low survivorship bias): **SPY** (US equity), **IEF** (7–10y Treasuries), **TLT** (20y+ Treasuries), **GLD** (gold), **DBC** (broad commodities).

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2. Covariance estimation (EWMA / RiskMetrics)

The strategy is driven by **second moments**, which are more forecastable than first moments. The default estimator is an exponentially-weighted (RiskMetrics-style) covariance with half-life H.

Decay per step and normalized weights over a trailing window of length T:

$$\lambda = 0.5^{1/H}, \qquad \tilde{w}_j = \frac{\lambda^{T-j}}{\sum_{k=1}^{T} \lambda^{T-k}}, \quad j = 1, \ldots, T.$$

Weighted mean and covariance:

$$\hat{\mu} = \sum_j \tilde{w}_j R_{j,\cdot}, \qquad \hat{\Sigma} = \sum_j \tilde{w}_j (R_{j,\cdot} - \hat{\mu})^\top (R_{j,\cdot} - \hat{\mu}),$$

symmetrized as $\frac{1}{2}(\hat{\Sigma} + \hat{\Sigma}^\top)$. The spine uses $H = 21$ (≈ a pandas `span`-60 EWMA), matching the horizon at which the strategy was validated.

Also available (`moments.jl`): unbiased `sample_cov`, Ledoit–Wolf constant-correlation `shrinkage_cov` (analytic optimal intensity), `nearest_psd`, `cov2cor`.

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3. The spine — portfolio construction

The spine is a **two-sleeve** book: a long-biased *base* that harvests the diversification premium by risk structure, plus a *trend* sleeve that is the divergent, crisis-hedging component. Each sleeve is independently volatility-targeted, the two are blended, and a regime overlay scales gross exposure.

3.1 Base sleeve — risk-based weights (no return forecast)

The base allocates by **risk**, not expected return. Three options (`riskbased.jl`), default **inverse-vol**:

- **Inverse-volatility** (naive risk parity):

$$w_i^{\text{iv}} = \frac{1/\sigma_i}{\sum_j 1/\sigma_j}, \qquad \sigma_i = \sqrt{\hat{\Sigma}_{ii}}.$$

- **Inverse-variance:** $w_i \propto 1/\sigma_i^2$.
- **Equal-risk-contribution (ERC / risk parity):** weights such that every asset contributes equally to portfolio variance. With marginal contribution $\mathrm{RC}_i = w_i(\hat{\Sigma}w)_i$, ERC solves the convex program

$$\min_{w>0}\ \tfrac{1}{2}\, w^\top \hat{\Sigma}\, w \ -\ \sum_i b_i \ln w_i, \qquad b_i = \tfrac{1}{N},$$

whose first-order condition is $w_i(\hat{\Sigma}w)_i = b_i$ for all i. It is solved by cyclical coordinate descent (solver-free, robust inside a backtest loop): for each i, solve the scalar quadratic

$$\hat{\Sigma}_{ii}\, w_i^2 + \Big(\textstyle\sum_{j\neq i} \hat{\Sigma}_{ij} w_j\Big) w_i - b_i = 0, \quad w_i > 0,$$

then renormalize $w \leftarrow w/\sum_i w_i$.

Also implemented: **maximum diversification** (Choueifaty–Coignard: maximize the diversification ratio $\frac{w^\top \sigma}{\sqrt{w^\top \hat{\Sigma} w}}$) and **Hierarchical Risk Parity** (López de Prado: correlation-distance $d_{ij} = \sqrt{(1-\rho_{ij})/2}$, hierarchical clustering, quasi-diagonalization, recursive bisection by inverse cluster variance — no matrix inversion, stable when N is large relative to sample length).

3.2 Trend sleeve — time-series momentum

The trend sleeve is the only place a *directional* view enters, and only at a horizon where momentum is real (§8.1). For each asset, the 12-month time-series-momentum sign:

$$s_i = \operatorname{sign}\Big(\textstyle\prod_{u=t-k+1}^{t} (1 + r_{u,i}) - 1\Big), \qquad k = 252.$$

Inverse-vol scaled and gross-normalized to a long/short book:

$$w_i^{\text{trend}} = \frac{s_i/\sigma_i}{\sum_j |s_j/\sigma_j|}, \qquad \textstyle\sum_i |w_i^{\text{trend}}| = 1.$$

This is the standard managed-futures construction (Moskowitz–Ooi–Pedersen): it goes long assets trending up and **short** assets trending down, sized to equalize each position's risk.

3.3 Per-sleeve volatility targeting (load-bearing)

Each sleeve is scaled to a target volatility $\sigma^\star$ (default 8%), using the sleeve's **own realized P&L volatility**, tracked with a RiskMetrics recursion on the realized sleeve return $p_t = w_{t-1} \cdot r_t$:

$$v_t^2 = \lambda\, v_{t-1}^2 + (1-\lambda)\, p_t^2, \qquad \hat{\sigma}_t^{\text{sleeve}} = \sqrt{v_t^2 \cdot \text{ppy}}.$$

Exposure multiplier (capped, de-risk-only when $\text{cap} \le 1$):

$$e_t = \min\Big(\text{cap},\ \frac{\sigma^\star}{\hat{\sigma}_t^{\text{sleeve}}}\Big).$$

> **Why realized-P&L vol, not ex-ante asset vol?** A long/short trend book's P&L can be *smooth while it works* (e.g. steadily short bonds in 2022) even as the underlying asset vols spike. Targeting on ex-ante asset-covariance vol would throttle the hedge exactly when it pays; targeting on the sleeve's own realized P&L keeps it engaged. This distinction is verified — using ex-ante vol collapses the strategy's crisis performance.

3.4 Blend and regime overlay

Blend the vol-targeted sleeves (default $\theta = 0.5$):

$$\tilde{w}_t = \theta\, e_t^{\text{base}}\, w_t^{\text{base}} \;+\; (1-\theta)\, e_t^{\text{trend}}\, w_t^{\text{trend}}.$$

A discrete **regime brake** scales gross exposure using the drawdown state of an equal-weight market proxy $m_t = \frac{1}{N}\sum_i r_{t,i}$ with index $I_t = \prod_{u\le t}(1+m_u)$:

$$\text{dd}_t = \frac{I_t}{\max_{u\le t} I_u} - 1, \qquad g_t = \begin{cases} \phi & \text{if } \text{dd}_t < -0.08 \\ 1 & \text{otherwise} \end{cases} \qquad (\phi = 0.5).$$

Final target book:

$$\boxed{w_t = g_t \cdot \tilde{w}_t\,.}$$

The regime brake was selected empirically over four candidate signals (vol-spike, drawdown, correlation, trend); the drawdown variant improved risk-adjusted return at ≈ zero cost, while the others whipsawed or never triggered on a diversified book (§8.2).

3.5 Stateful daily update

In production the spine is stateful: `SpineState` carries each sleeve's RiskMetrics variance ($v^{\text{base}}, v^{\text{trend}}$) and the last-held weights across days. Each trading day, `spine_step!(state, window)`:

1. folds the just-realized bar into each sleeve's vol via §3.3 (using the previously-held weights);
2. recomputes base and trend weights from the trailing window (§3.1–3.2);
3. sizes each sleeve (§3.3), blends (§3.4), applies the regime brake;
4. returns the target book and stores the new sleeve weights.

Positions are then converted to signed share targets $q_i = w_i \cdot C/P_i$ (capital C, price P_i) and reconciled against the broker so the strategy trades the **delta**, not the full target, each day.

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4. Risk metrics

Standard definitions (`metrics.jl`), all annualized with $\text{ppy} = 252$.

- **Annualized return** (geometric): $\left(\prod_t (1+r_t)\right)^{\mathrm{ppy}/T} - 1$.
- **Annualized volatility:** $\mathrm{std}(r)\sqrt{\mathrm{ppy}}$.
- **Sharpe ratio:** $\dfrac{\bar{r}\,\mathrm{ppy} - r_f}{\mathrm{std}(r)\sqrt{\mathrm{ppy}}}$.
- **Sortino ratio** (downside deviation about a minimum acceptable return MAR):

$$\mathrm{DD} = \sqrt{\mathrm{mean}\left(\min(r - \mathrm{MAR}, 0)^2\right)}\,\sqrt{\mathrm{ppy}}, \qquad \mathrm{Sortino} = \frac{\mathrm{AnnRet} - r_f}{\mathrm{DD}}.$$

- **Drawdown series and max drawdown:** with cumulative $C_t = \prod_{u \le t}(1 + r_u)$,

$$\mathrm{DD}_t = \frac{C_t}{\max_{u \le t} C_u} - 1, \qquad \mathrm{MaxDD} = \min_t \mathrm{DD}_t.$$

- **Calmar ratio:** $\mathrm{AnnRet} \,/\, |\mathrm{MaxDD}|$.
- **Value-at-Risk** at confidence α (three methods):
  - *Historical:* $\mathrm{VaR}_\alpha = -Q_{1-\alpha}(r)$.
  - *Gaussian:* $-(\mu + z_{1-\alpha}\sigma)$.
  - *Cornish–Fisher* (skew/excess-kurtosis adjusted):

$$z_{cf} = z + (z^2 - 1)\tfrac{s}{6} + (z^3 - 3z)\tfrac{k}{24} - (2z^3 - 5z)\tfrac{s^2}{36}, \qquad \mathrm{VaR}_\alpha = -(\mu + z_{cf}\,\sigma).$$

- **Expected shortfall (CVaR):** $\mathrm{ES}_\alpha = -\,\mathrm{mean}\{\, r : r \le Q_{1-\alpha}(r) \,\}$.

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5. Governed execution — risk invariants as guards

Every order passes through one governed path (`ExecutionController`) that enforces hard, testable invariants before the order can reach a venue. These are risk *controls*, not strategy — the strategy proposes, the controller disposes.

INVARIANT	RULE
Idempotency (REQ-EXEC-002)	Client order id = `pool|symbol|solve_id`; a repeated solve never double-submits (also deduped broker-side).
Per-pool budget (REQ-RISK-003)	Reject if cumulative emitted notional in a pool exceeds its daily budget; reset each trading day.
Daily loss halt (REQ-RISK-004)	Halt the pool if realized daily P&L $\leq -L_{\max}$.
Data staleness (REQ-DATA-003)	Reject emission if the pool's market data is older than a threshold.
Reconciliation (REQ-EXEC-003)	Compare fill-driven expected positions to broker positions; on divergence beyond tolerance, halt.
Kill switch (REQ-GOV-002)	A manual/alert stop halts all normal emission; liquidation orders bypass it (you must always be able to flatten).
Fill lineage (REQ-AUDIT-001)	Every recorded fill carries `signal_id / regime / solve_id / order_id`.

5.1 Layer-3 pre-trade safety gate (live money)

Before any order in live mode, a pre-flight aggregate must pass (`SafetyGate.preflight`), else the book is halted and an alert fires. Guards include:

- **Drawdown halt:** with a persisted equity high-water mark HWM, halt if $\frac{E}{\mathrm{HWM}} - 1 < -d_{\max}$ (default 15%).
- **Daily-loss halt:** halt if $E - E_{\text{prev}} < -L_{\max}$.
- **Gross-leverage cap:** reject if $\frac{\sum_i |q_i| P_i}{E} > \Lambda_{\max}$ (default 2×).
- **Per-name cap:** reject if any $|q_i| P_i / E$ exceeds a bound (default 0.85 — a *runaway-bug* catcher; genuine concentration risk is handled by the risk-based construction and the gross/drawdown caps, since inverse-vol legitimately places large weight in low-vol bonds).
- **Account & data checks:** account active/not blocked, sufficient buying power, data fresh.

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6. Composition — the daily pipeline

$$\text{prices} \xrightarrow{\S1} R \xrightarrow{\S2} \hat{\Sigma} \xrightarrow{\S3} \underbrace{w_t}_{\text{spine target}} \xrightarrow{\times C/P} q_t \xrightarrow[\text{seed from broker}]{\text{delta}} \underbrace{\text{governed orders}}_{\S5} \to \text{fills} \to \text{reconcile.}$$

One rebalance per trading day (§8.3 explains why not intraday), executed pre-open so orders fill at the open.

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7. Validation methodology — measuring edge honestly

The platform's design was gated on **out-of-sample edge tests**, not in-sample fit.

7.1 Information Coefficient

For a per-bar cross-sectional score $x_{t,i}$ and forward return $r_{t+1,i}$, the **Information Coefficient** is the Spearman rank correlation across assets each window; the **IC information ratio** is its t-statistic across N_w windows:

$$\mathrm{IC}_t = \rho_{\mathrm{Spearman}}\big(x_{t,\cdot},\, r_{t+1,\cdot}\big), \qquad \text{IC-IR} = \frac{\overline{\mathrm{IC}}}{\mathrm{std}(\mathrm{IC})}\sqrt{N_w}.$$

A signal is treated as real only if roughly $|\overline{\mathrm{IC}}| > 0.02$ **and** $|\text{IC-IR}| > 2$.

7.2 Out-of-sample discipline

- **Train/test split** and walk-forward evaluation; parameters fixed *a priori* (round numbers), never tuned on the test window.
- **Control for confounds:** e.g. when testing a regime-conditional blend, compare not only to the static baseline but to *static-at-the-adaptive-rule's-own-mean-weight*, to separate genuine timing value from "just ran a different constant."
- **Purged K-fold / combinatorial-purged CV** (López de Prado) is available for series with overlapping labels (`module_11_cv`).
- **Multiple-testing awareness:** a rule is granted one or two well-motivated shots; fishing variants until one passes is treated as p-hacking and rejected.

7.3 The decisive result

On 60-day 15-minute data (≈1,547 bars × 60 S&P names), two candidate alpha models were tested:

MODEL	DIRECTIONAL ACCURACY	p VS 50%	IC
3-factor cross-sectional	50.9%	0.48	−0.004
Bayesian (Kalman/NIG/James–Stein/ADVI)	50.7%	0.59	+0.001

Both are **coin flips.** There was no exploitable predictive edge at this horizon/universe. This is *why* the platform harvests risk premia rather than forecasts returns.

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8. Empirical findings that shaped the design

All figures are historical backtests, net of ~2 bps/side costs, on the reference universe.

8.1 Momentum vs. reversal by horizon

IC of a trailing k-day return against the next-day return, pooled across assets, 2006–2026:

k	1D	2D	3D	5D	10D	21D	63D	126D	252D
IC	−.038	**−.041**	−.035	−.031	−.021	−.010	−.000	+.005	+.002

Short horizons (1–63d) exhibit **reversal** (a sharp drop tends to bounce); **momentum** appears only at 126–252d. This is the quantitative reason the trend sleeve uses a **12-month** lookback and why "shorting a 2-day selloff" fights the strongest short-horizon force in the data.

8.2 The validated spine

Base ⊕ trend ⊕ vol-target ⊕ regime brake, 2007–2026 (incl. GFC, COVID, 2022):

CONFIGURATION	SHARPE	SORTINO	MAX DD	2022
Diversified base only	0.85	1.24	−18%	−15%
Spine (base + trend), regime `:none`	**0.94**	1.28	**−10.6%**	−3.2%
Spine, regime `:dd` (production)	**0.97**	1.30	−10.6%	**−1.4%**

The trend sleeve made **+4.4% in 2022** (short bonds / long energy) while a long-only book bled — that is the crisis hedge working as designed.

8.3 Cadence — daily, not intraday

On 17,659 real 15-minute bars (2022–2026), applying the daily book intraday: holding through the day gives Sharpe ≈ 1.24. An intraday vol-brake **never triggered** (a 5-asset diversified book barely moves intraday — stocks/bonds/gold/commodities offset within the bar; diversification *is* the intraday shock absorber). An intraday drawdown-brake **whipsawed** (−1% CAGR for −0.7% max-DD). Full 15-minute re-optimization is pure cost drag — risk premia do not change in 15 minutes. **Conclusion: once-daily rebalance.**

8.4 Leverage — the return/drawdown trade-off

Leverage L on the strategy return r_t, financing the borrowed portion at annual rate f:

$$r_t^L = L\, r_t - (L-1)\,\frac{f}{\mathrm{ppy}}.$$

2007–2026, financing at 3% (the friendliest case) vs. 6.5% (today's retail margin):

L	CAGR @3%	CAGR @6.5%	VOL	SHARPE	MAX DD
1.0	+5.9%	+5.9%	6.1%	0.97	−10.6%
1.5	+7.2%	+5.4%	9.2%	0.81	−18.1%
2.0	+8.5%	+4.7%	12.2%	0.73	−25.1%
3.0	+10.6%	+3.2%	18.3%	0.64	−39.1%

Two facts: (i) **Sharpe only falls with leverage** (financing is a pure drag on risk-adjusted return); (ii) at today's ~6.5% margin, leverage is **net-negative** — the strategy's ~6% return is below the cost of borrowing, so leverage subtracts return while amplifying drawdown roughly $\propto L$. Double-digit CAGR requires ~3×, which comes with a **−39% drawdown**. There is no double-digit-return-at-low-risk configuration.

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9. The portfolio-optimization library — optimizers, Monte-Carlo robustification, and a service

The spine (§3) is one strategy built on a **general-purpose optimization library**. `module_13_portfolio` (`PortfolioOpt`) is usable on its own: every optimizer consumes the moments/covariance of §1–2 (or a raw return matrix) and returns budget-constrained weights. Convex programs are solved with **JuMP** and the **Clarabel** conic solver; a long-only toggle and scalar/vector box bounds are shared across the family.

> **Caveat, stated up front.** Any optimizer that consumes expected returns μ (§9.1, §9.2) inherits the Markowitz *error-maximization* problem — it over-fits noise in the sample mean. That is *why* the flagship spine avoids μ entirely (§3), and why this library ships the antidotes next to the optimizers: shrinkage (§2), equilibrium anchoring (§9.2), and Monte-Carlo resampling (§9.5).

9.1 Markowitz family (`meanvariance.jl`)

Quadratic programs over $\sum_i w_i = 1$ with optional long-only and box constraints:

- **Minimum variance:** $\min_w \; w^\top \hat{\Sigma} w$ — the one member that uses no μ, hence the robust one.
- **Maximum Sharpe** (tangency): $\max_w \; \dfrac{\mu^\top w - r_f}{\sqrt{w^\top \hat{\Sigma} w}}$, via the standard convex reformulation.
- **Mean-variance:** $\max_w \; \mu^\top w - \frac{\gamma}{2} w^\top \hat{\Sigma} w$ (risk-aversion γ) or $\min_w \; w^\top \hat{\Sigma} w$ s.t. $\mu^\top w \geq r^\star$ (target-return).
- **Efficient frontier:** the target-return sweep across n points.

9.2 Black–Litterman (`blacklitterman.jl`)

Anchor on the market and blend in views, rather than trusting raw sample means. Reverse-optimize the equilibrium prior from market-cap weights,

$$\Pi = \delta\,\hat{\Sigma}\,w_{\mathrm{mkt}},$$

then form the posterior from K views (P, Q) with view-uncertainty Ω and prior scaling τ:

$$\mu_{\mathrm{BL}} = \Big[(\tau\hat{\Sigma})^{-1} + P^\top\Omega^{-1}P\Big]^{-1}\Big[(\tau\hat{\Sigma})^{-1}\Pi + P^\top\Omega^{-1}Q\Big], \qquad \Sigma_{\mathrm{BL}} = \hat{\Sigma} + \Big[(\tau\hat{\Sigma})^{-1} + P^\top\Omega^{-1}P\Big]^{-1}.$$

`make_view` builds a row of P with its Q entry ("these assets will return q"); the posterior $(\mu_{\mathrm{BL}}, \Sigma_{\mathrm{BL}})$ then feeds any §9.1 optimizer. Omitting Ω defaults it to $\mathbf{diag}(P\,\tau\hat{\Sigma}\,P^\top)$.

9.3 Tail-risk optimizers (`tailrisk.jl`)

These shape the *realized* loss distribution from the scenario matrix R (not a Gaussian proxy). **Minimum CVaR** uses the Rockafellar–Uryasev linear program: at level α,

$$\min_{w,\zeta,u}\ \zeta + \frac{1}{(1-\alpha)T}\sum_{t=1}^{T} u_t \quad \text{s.t.} \quad u_t \ge -R_{t,\cdot}\,w - \zeta,\ \ u_t \ge 0,\ \ \textstyle\sum_i w_i = 1,$$

optionally with a return floor $\mu^\top w \ge r^\star$. It remains an **LP** regardless of scenario count. **Minimum CDaR** is the drawdown analogue (conditional drawdown-at-risk).

9.4 Cost-aware optimization (`costaware.jl`)

Decide what to *trade to* given current holdings w_0, so cost shapes the solution instead of being measured afterward:

$$\text{turnover} = \sum_i |w_i - w_{0,i}|, \qquad \text{cost}(w_0 \to w) = c_{\mathrm{lin}}\sum_i |\Delta_i| \ + \ c_{\mathrm{imp}}\sum_i \Delta_i^2, \qquad \Delta_i = w_i - w_{0,i},$$

and `mean_variance_tc` maximizes $\mu^\top w - \frac{\gamma}{2} w^\top\hat{\Sigma}w - \text{cost}(w_0 \to w)$ — a linear (spread/commission) plus convex (market-impact) penalty *inside* the objective.

9.5 Monte-Carlo robustification (`robust.jl`)

This is the platform's Monte Carlo — used to **defend against estimation error**, not to forecast price. A data-generating process is a pluggable sampler `(R, T_sim) -> simulated returns` :

- `gaussian_dgp` — draw from $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$;
- `iid_bootstrap` — resample whole rows with replacement (nonparametric; preserves the cross-sectional dependence);
- `block_bootstrap(block)` — circular blocks (preserves short-horizon autocorrelation);
- `stationary_bootstrap(meanblock)` — Politis–Romano geometric block lengths;
- `student_t_dgp(ν)` — fat-tailed multivariate-t scaled to $\hat{\Sigma}$.

Two consumers:

- **Feasible-set Monte Carlo** (`random_portfolios`): sample many admissible weight vectors to trace the achievable risk/return cloud — the dashboard's scatter, and a sanity check on any optimizer's output.
- **Resampled (Michaud) efficient frontier** (`resampled_frontier`): for each of n_{resample} draws, simulate T_{sim} returns from the sampler, compute that draw's frontier, and **average the weights** across draws. The averaged frontier is markedly more stable out-of-sample than the single-shot Markowitz frontier — the direct antidote to §9.1's error-maximization.

9.6 The optimizer as a service (`scripts/`)

The whole stack is exposed over HTTP for a dashboard or an external caller — `portfolio_server.jl` (JSON REST, default `127.0.0.1:8766`) with a Python **Dash** frontend (`dashboard.py`):

ENDPOINT	RETURNS
`POST /optimize`	one strategy's weights (min-var, max-Sharpe, risk-parity, min-CVaR, ...)
`POST /frontier`	efficient frontier (n points)
`POST /resampled_frontier`	Michaud resampled frontier (§9.5)
`POST /backtest`	walk-forward backtest of a chosen strategy
`POST /metrics`	performance metrics (§4) for a return series
`POST /random_portfolios`	feasible-set Monte-Carlo cloud (§9.5)
`GET /health`	liveness

```
julia --project=. scripts/portfolio_server.jl     # backend on :8766
python scripts/dashboard.py                        # Dash UI on :8050
```

This makes the repository usable as a **general portfolio-optimization library and service**, with the spine as the flagship strategy built on top of it. (A production data layer — `module_1_data/data_feeds_production.jl` — can feed it live inputs including a **Deribit BTC volatility-index** signal; crypto enters as a risk input, not as a traded sleeve.)

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10. The research track (Python) — archived, and why

The platform began as an alpha-prediction system. That code is preserved (archived) because the mathematics is sound, but it is **not on the live path** — edge validation (§7) found no predictive signal at the horizons tested. Documented honestly:

- **Bayesian alpha engine** (`signal_engine.py`): per-ticker return estimation via Kalman filtering → Normal-Inverse-Gamma conjugate updating → James–Stein shrinkage → ADVI, combined by inverse-variance weighting. Mathematically correct; **no measured predictive edge** (§7.3).
- **Monte-Carlo risk engine** (`monte_carlo_engine.py`): geometric-Brownian-motion simulation of a portfolio given *assumed* drift and volatility, producing VaR/CVaR/expected-shortfall/probability-of-profit. It is a **risk calculator, not an optimizer** — it sizes a position's tail; it does not create return. Correct tool for sizing a premium by its tail; wrong tool to find one.
- **Six-Sigma Oracle** (`risk_engine.py`): samples weight vectors and minimizes a simulated loss rate subject to a return target — i.e. Sharpe maximization on **historical means**. This is the Markowitz *error-maximization* trap: optimizing on noisy sample means fails out-of-sample. Its correct use is **survival-sizing** ("at exposure X, what is the six-sigma loss?" → cap gross so the tail stays within budget), not return generation.

The lesson, stated once: sophisticated machinery (long/short, QP optimization, Monte-Carlo, Bayesian filtering) *packages and sizes* a signal; it does not *create* one. The signal is the constraint, and at the tested horizons it was not there.

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11. Code map

CONCEPT (§)	FILE · KEY SYMBOLS
Returns, moments, EWMA cov (§1–2)	`src/module_13_portfolio/moments.jl` · `simple_returns`, `ewma_cov`, `shrinkage_cov`, `nearest_psd`
Base weights (§3.1)	`src/module_13_portfolio/riskbased.jl` · `inverse_variance`, `risk_parity`, `max_diversification`, `hrp_weights`
Spine: trend, vol-target, blend, regime, stateful (§3.2–3.5)	`src/module_13_portfolio/spine.jl` · `tsmom_signal`, `tsmom_weights`, `voltarget_exposure`, `regime_multiplier`, `SpineState`, `spine_step!`, `spine_targets`
Metrics, VaR/CVaR (§4)	`src/module_13_portfolio/metrics.jl` · `sharpe`, `sortino`, `max_drawdown`, `calmar`, `value_at_risk`, `expected_shortfall`
Optimizer library — Markowitz / BL / tail-risk / cost-aware (§9.1–9.4)	`meanvariance.jl` · `min_variance`, `max_sharpe`, `mean_variance`, `efficient_frontier`; `blacklitterman.jl` · `black_litterman`, `implied_equilibrium_returns`, `make_view`; `tailrisk.jl` · `min_cvar`, `min_cdar`; `costaware.jl` · `mean_variance_tc`, `transaction_cost`, `turnover`
Monte-Carlo robustification (§9.5)	`src/module_13_portfolio/robust.jl` · `gaussian_dgp`, `iid_bootstrap`, `block_bootstrap`, `stationary_bootstrap`, `student_t_dgp`, `random_portfolios`, `resampled_frontier`
Optimizer REST service + dashboard (§9.6)	`scripts/portfolio_server.jl`, `scripts/dashboard.py`
Production data feeds (Cboe / FRED / Deribit crypto-vol)	`src/module_1_data/data_feeds_production.jl`
Purged / combinatorial CV (§7.2)	`src/module_11_cv/`
Governed execution invariants (§5)	`src/module_7_execution/` · `execution_controller.jl`, `venue_interface.jl`, `venues/`
Layer-3 safety gate (§5.1)	`src/module_8_governance/safety_gate.jl` · `SafetyLimits`, `preflight`, `drawdown`
Data adapters (Alpaca / IBKR / CSV)	`src/module_1_data/` · `equity_panel.jl`, `alpaca_panel.jl`, `ibkr_panel.jl`
Live driver + safety	`scripts/spine_live.jl`, `scripts/run_spine_daily.sh`
Leverage / cadence / validation analyses (§7–8)	`scripts/leverage_decision_data.jl`, `docs/CANONICAL_ARCHITECTURE.md`, `docs/leverage_decision.html`

CONCEPT (§)	FILE · KEY SYMBOLS
Research track (archived)	`Archive/blaque_baux/…` · `signal_engine.py`, `optimizer/risk_engine.py`; `Archive/crypto-quant-mvp-v2/monte_carlo_engine.py`

12. References

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13. Disclaimer

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